

Mail server security forensic report

SESSIONS
INVESTIGATED

19

VULNERABLE

7

FLAGGED

2

SECURE

10

KEY FINDINGS

- 13 of 19 sessions were encrypted; 6 were plaintext.
- 5 sessions exposed login credentials in plaintext.
- 3 encrypted sessions had a certificate that failed validation.
- 1 session showed a STARTTLS downgrade (encryption requested but never started).

SESSION DETAILS

Session	Proto	TLS	Enc.	Creds	Cert	Verdict	Reason
bad_capture1.pcap	SMTP	no TLS	no	yes	n/a	VULNERABLE 100	Credentials exposed in plaintext
bad_capture2.pcap	SMTP	no TLS	no	yes	n/a	VULNERABLE 100	Credentials exposed in plaintext
bad_capture3.pcap	SMTP	no TLS	no	yes	n/a	VULNERABLE 100	Credentials exposed in plaintext
bad_capture4_imap.pcap	IMAP	no TLS	no	yes	n/a	VULNERABLE 100	Credentials exposed in plaintext
bad_capture5_pop3.pcap	POP3	no TLS	no	yes	n/a	VULNERABLE 100	Credentials exposed in plaintext
good_capture5_downgrade.pcap	SMTP	no TLS	no	no	n/a	VULNERABLE 100	STARTTLS requested but no TLS handshake found – downgrade
good_capture8_weaktls.pcap	SMTP	TLS (old)	yes	no	no	VULNERABLE 100	Old or weak TLS in use.
good_capture11_expiredcert.pcap	SMTP	TLS 1.2	yes	no	no	FLAGGED 50	Certificate failed validation (for example expired or not trusted).
good_capture14_tls12expired.pcap	SMTP	TLS 1.2	yes	no	no	FLAGGED 50	Chain contains expired certificate: CN=mail.good.test
good_capture1.pcap	SMTP	TLS 1.2	yes	no	yes	SECURE 0	

Session	Proto	TLS	Enc.	Creds	Cert	Verdict	Reason
good_capture10_pop3.pcap	POP3	TLS 1.2	yes	no	yes	SECURE 0	
good_capture12_certchain.pcap	SMTP	TLS 1.2	yes	no	yes	SECURE 0	
good_capture13_tls12valid.pcap	SMTP	TLS 1.2	yes	no	yes	SECURE 0	
good_capture2.pcap	SMTP	TLS 1.2	yes	no	yes	SECURE 0	
good_capture3.pcap	SMTP	TLS 1.2	yes	no	yes	SECURE 0	
good_capture4_attachment.pcap	SMTP	TLS 1.2	yes	no	yes	SECURE 0	
good_capture6_multirecipient.pcap	SMTP	TLS 1.2	yes	no	yes	SECURE 0	
good_capture7_largeattachment.pcap	SMTP	TLS 1.2	yes	no	yes	SECURE 0	
good_capture9_imap.pcap	IMAP	TLS 1.2	yes	no	yes	SECURE 0	

Method: passive analysis of packet captures (PCAP) of SMTP, IMAP and POP3 traffic. Source: real_data.csv snapshot built into this page.